Concept Review

Brushless DC (BLDC) Motors

Why use BLDC Motors?

Brushless DC (BLDC) motors are efficient and more durable than regular DC motors. Instead of using brushes or a commutator to deliver current to the coils, BLDC motors use a rotating permanent magnet and stationary coils. This design reduces wear and electrical noise while still allowing precise control of torque and speed. By using electronic switching of current, BLDC motors are able to maintain high torque throughout rotation, making them highly power efficient. BLDC's efficiency and reliability make them ideal for applications that require precision and continuous operation, such as washing machines, air conditioners, vacuums, and hard disc drives.

Introduction

A brushless DC (BLDC) motor is an electric motor that uses electronic commutation instead of brushes to switch the direction of current. It generates torque by placing permanent magnets in the rotor and energizing electromagnets in the stator. Unlike brushed motors, BLDC motors do not require brushes to deliver current to the rotor, resulting in longer lifespan and lower maintenance since the mechanical elements are not in direct contact and are therefore less likely to wear overtime. The design of a brushless DC motor is more complicated than that of a brushed DC motor, as additional sensors are needed to detect the angular position of the armature and more complex circuitry and feedback control is necessary to ensure proper commutation of the electromagnetic fields in the stator, which increases the initial cost. Despite this, their durability and efficiency make them ideal for various industrial applications.

Basic Construction and Operation

A BLDC motor has 2 main components: a **rotor**, which has a permanent magnet, and as the name suggest, is the part that rotates, and a **stator**, which has windings to create the magnetic field that rotates the rotor.

The way the rotor's magnets interact with the electromagnetic field generated by the stator is how the motor rotates. As magnets, they attract opposite poles and repel the same poles. In a simple DC motor, this is a mechanical process triggered by a commutator with brushes. In a BLDC motor, it happens electronically with the help of transistor switches.

BLDCs constructed with the rotor located inside the stator are called **inrunner** motors. Inrunners have a more lightweight construction and higher rotational speeds because of their smaller rotating diameter. BLDCs that have the rotor spinning outside the fixed inner stator are called **outrunner** motors. Outrunner motors have a higher torque because of the larger rotor diameter and the resulting force applied further away from the center (larger moment arm). These two configurations are shown in Figure 1.

Windings

The stator comes in two winding configurations. A **Delta configuration** connects the windings in a triangular shape, where two windings are connected to a common node as shown in Figure 1. This configuration provides low torque at low speeds but allows for a higher top speed. Conversely, the **Wye configuration** has all three windings meeting at the center, forming a Y shape, as shown in Figure 1. It provides high torque at low speeds and a lower top speed.

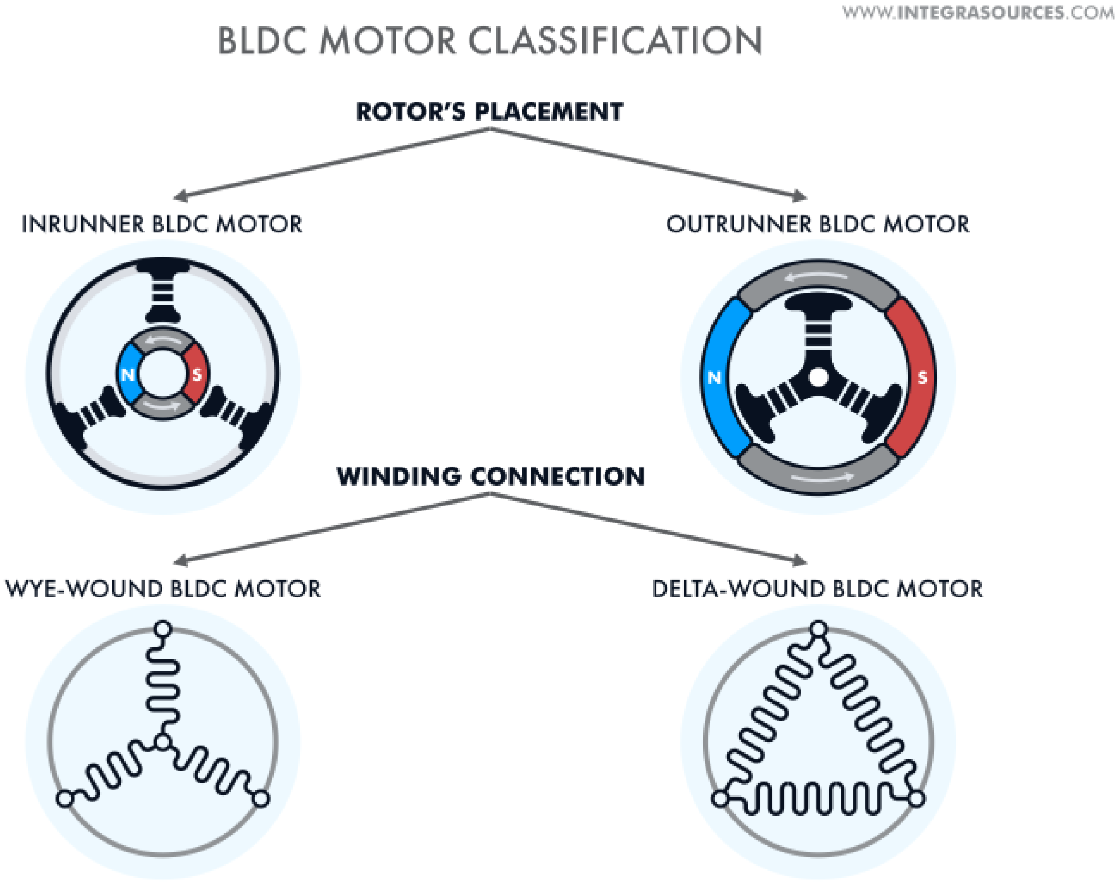


Figure 1. BLDC Motor Classification by Rotor Placement and Winding Configuration

Measurement

The motor controller measures the angular position of the rotor through sensors, such as rotary encoders or Hall Effect, or through sensorless techniques using back EMF. With this information, a motor controller generates alternating current (AC) signals that individually drive the electromagnets in the stator. This makes brushless DC motors more efficient than brushed DC motors since the applied voltage is pulsed as opposed to direct. In this context, AC does not imply a sinusoidal waveform but rather bi-directional current that can follow an arbitrary waveform.

The brushless DC motor on the Mechatronic Actuators application board has built-in Hall effect sensors, which are transducers that vary their output voltage in response to a magnetic field. These kinds of sensors are most frequently used for proximity switching and current sensing, as well as position and speed control of brushless DC motors. These sensors, unlike sensorless techniques, allow for accurate positioning feedback of the BLDC without needing the motor to start moving.

In a typical configuration, there are three Hall effect sensors separated by 120˚ placed around the stator that can detect the position of the armature, similar to Figure 2. There is a total of six possible combinations of active/inactive Hall effect sensor states before the sensing pattern is repeated. For every 60˚ rotation, one of the Hall sensors changes its state; it takes six steps to complete a whole electrical cycle. This is then used to determine the commutation sequence. In each step, there is one motor terminal driven high, another motor terminal driven low, with the third one left floating. Note that the other two theoretical combinations of sensors (all high or all low) can never be reached and represent a faulty state.

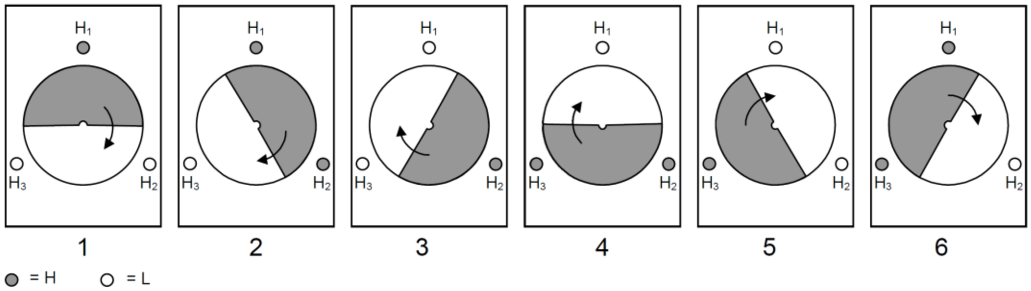
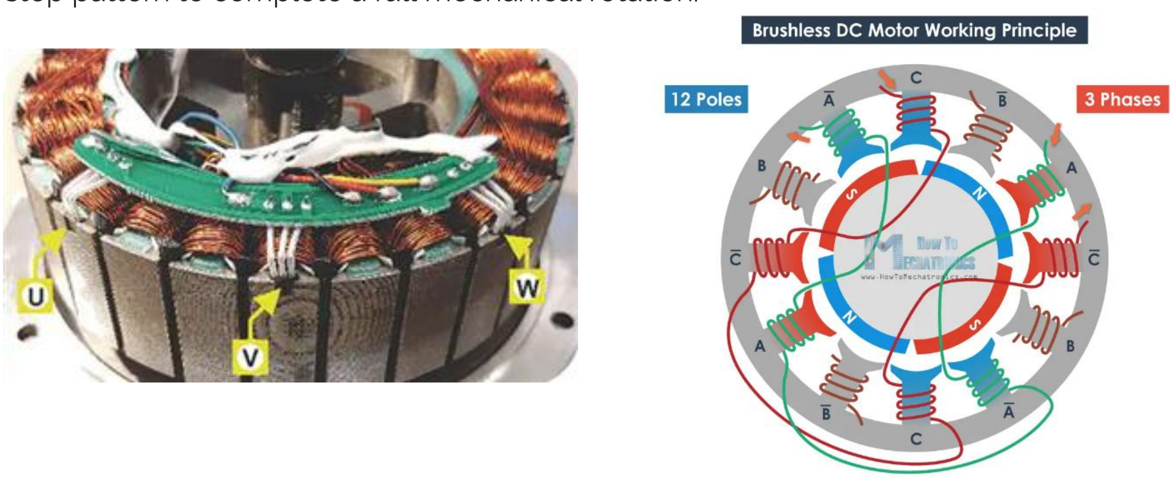


Figure 2. Hall Sensor Reading in a BLDC Signal Cycle

This diagram, as well as the ones in the following sections, are an equivalent but simplified versions of how the inside of a BLDC motor works. Most motors will have the hall sensors closer together as shown in Figure 3a, due to the coils position (Figure 3b), however, they create an equivalent output as if the sensors were placed 120˚ away from each other.

A signal cycle, the 6-step pattern, does not always correspond to a full mechanical revolution of the BLDC motor. The number of cycles to complete a mechanical rotation is equal to the amount of rotor pole pairs (not rotor poles). These poles are the coils seen in Figure 3. For a 2-pole pair BLDC motor, the signal cycle has to be repeated two times for a full mechanical rotation; in other words, one signal cycle is only half of the rotation of the rotor. In the case of Figure 3b, there are 12 poles (coils) and therefore 6 pairs, which would require 6 cycles of the 6 step pattern to complete a full mechanical rotation.



a) Position of hall sensors

b) Poles in a BLDC motor

Figure 3. Insides of a BLDC motor

Driving a BLDC Motor

The BLDC motor supplied with the Actuators Trainer has stator coils wired in a Wye configuration. To control the BLDC motor, one can use the phase/Hall effect sensor diagram in Figure 2.

When energizing a coil, a magnetic field is generated that is 90˚ counterclockwise from the direction of current in the winding, as shown in Figure 4 as a white arrow. Energizing a coil would move the magnet's north to follow this magnetic field. Note that the figure has coils named A, B, and C, and the hall sensors are denoted with shapes: a circle, triangle and square.

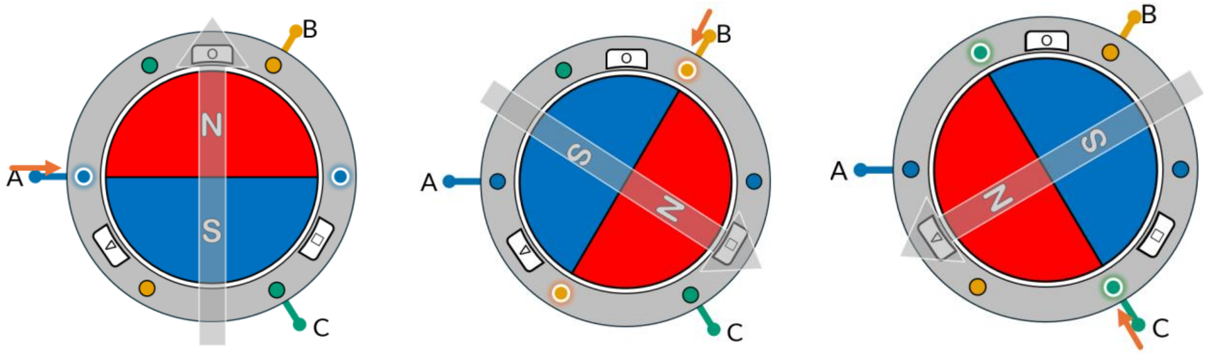


Figure 4. Magnet positions and magnetic fields when energizing coils

However, due to the Y configuration of the coils, current needs to flow from one set of coils to the other, for example, from A to C.

The desired states can be achieved using three half H-bridges in the circuit, which can be visualized in the simplified diagram using switches in Figure 5. Note that if you were to energize a single coil, the other two would act as ground and the current would be divided between both of them, which is what would be theoretically happening in Figure 4. This is not used when driving a motor but can help with understanding them.

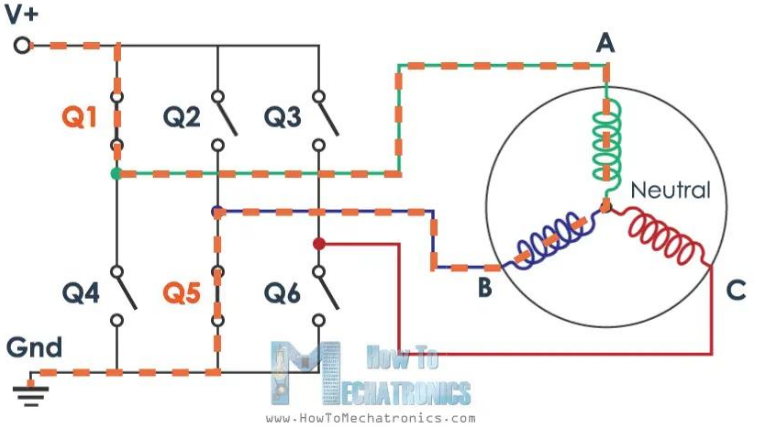


Figure 5. Three half H-bridges required to drive the brushless DC motor

This means that there are 6 available commutating steps as shown in Figure 6. These 6 steps match the 6 steps of the Hall sensors. When the position of the rotor changes, as measured by the hall sensors, that means that the motor has arrived at a new position and a magnetic field needs to be generated to attract the rotor's permanent magnet to the next position. That is why we activate each coil, one after another, so that it keeps rotating, therefore trying to align the permanent magnets to the magnetic field from the coils constantly.

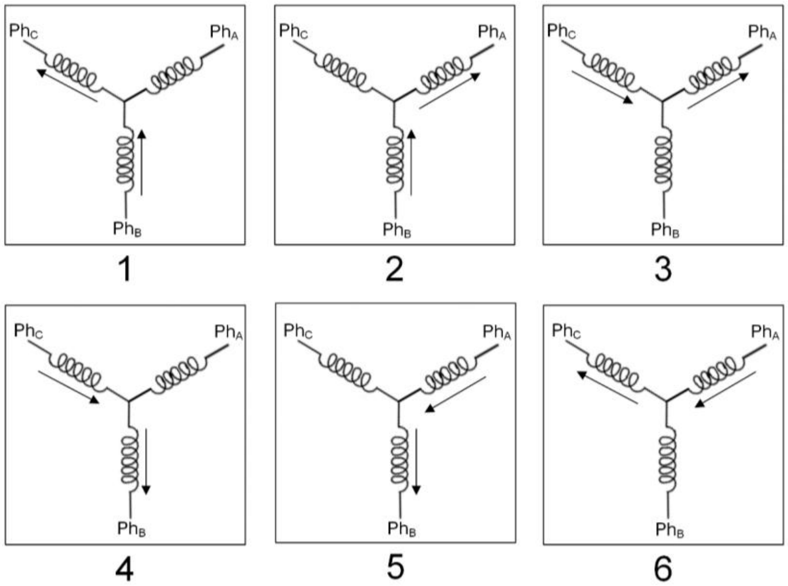


Figure 6. Six Step Commutation

Note that to achieve maximum torque, the angle between the magnetic field of the stator and the rotor should be ±90° depending on the desired directionality as shown in Figure 7.

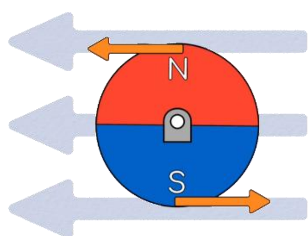
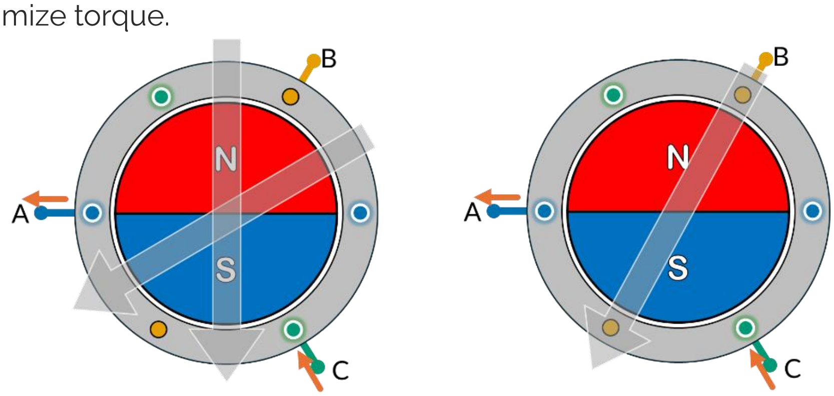


Figure 7. Magnetic Field for Maximum Torque

Below (Figure 8) are 3 of the 6 available commutation steps and the magnetic fields they each generate. The 3 missing steps are the inverse on where the current flows in and out of the circuit, which would create just a magnetic field in the opposite direction of the one shown in the picture. At each of the 6 values read by the hall sensors, the controller should energize the coils in a way that creates a magnetic field with a 90-degree offset towards the desired rotation to maximize torque.



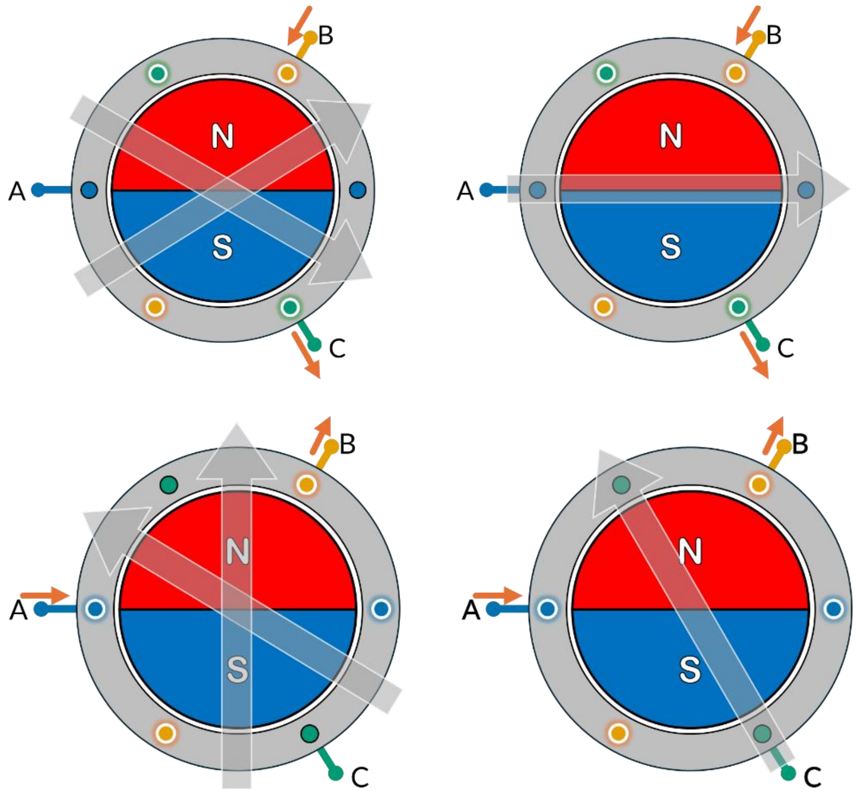


Figure 8. Magnetic Fields from Energizing 2 Coils and Equivalent Magnetic Field

The energizing sequence to rotate the motor would look something like Figure 9,where a, b, c are hall sensor outputs and U, V, W are the coils. For each different Hall sensor reading, there is a different commutation step in the coils, switching between high (current going in), float (disconnected), and low (current going out).

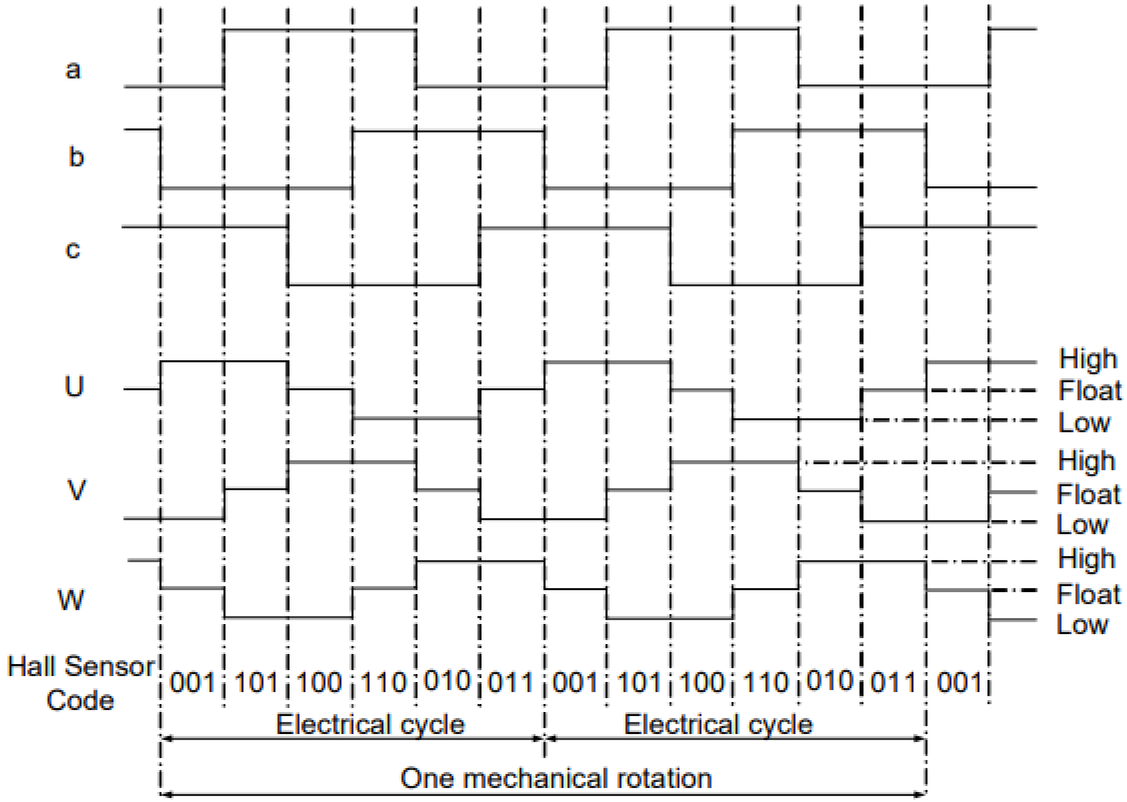


Figure 9. Three-phase BLDC motor sensor versus drive timing

Datasheet And Considerations

The motor's datasheet summarizes its key electrical and mechanical characteristics and are essential when selecting a motor for a project or application.

Understanding a BLDC Motor Datasheet

When selecting a BLDC motor for a project, it is helpful to understand each key specification from the datasheet. This section will mostly focus on the electrical information, however, note that there will also be mechanical information about the motor, usually in the form of a mechanical drawing. This information is important for deciding if a motor will physically fit in a required space, or mount effectively. It is important to also consider whether the motor is sensored or sensorless.

Rated Voltage

This is the nominal voltage at which the motor operates most efficiently. Operating at or near this voltage ensures stable performance without overheating or loss of torque. If the supply voltage is too low, the motor may run sluggishly; too high, and it risks damage. Always select a motor with a rated voltage close to your available power source—whether it's a battery or a regulated DC supply.

Kv Rating (RPM/V)

The Kv rating tells you how fast the motor spins for each applied volt with no load. For example, a 1000 Kv motor at 12V would theoretically spin at 12,000 RPM. Higher Kv motors are designed for speed but deliver less torque, while lower Kv motors spin slower but provide more torque. This makes Kv one of the most important specifications to match with your application's torque-speed trade-off—think high Kv for drones, and low Kv for robotic joints or wheel drives. However, in many cases, the rated no-load maximum speed of the motor will be less useful than the torque-speed curve, which some motors have available.

Max Continuous Current / Rated Current (A)

This is the highest current the motor can safely draw continuously without overheating. It's tied directly to how much torque the motor can generate over time. The motor draws the most current when exerting maximum torque, so when it is stalled. Choosing a motor with a higher rated current gives you more sustained torque, but also means you'll need a power supply and driver capable of delivering it. Staying within this limit is crucial for long-term reliability, especially if your application involves frequent or sustained high loads.

Stall Current and Stall Torque

Stall/peak torque is the maximum torque the motor can produce at zero speed, and stall current is the very high current it draws under these conditions. While these values are useful for estimating startup behavior (e.g., a robot lifting an arm from rest), motors should not operate at stall for more than a few seconds or they'll quickly overheat. Consider these specs for applications with high starting loads or frequent stops and starts.

Torque Constant (Kt)

The torque constant (usually in Nm/A) defines how much torque the motor produces per amp of current. A higher Kt means more torque with less current, which is helpful in low-voltage systems or when thermal efficiency is important. This is particularly valuable in precision applications like servo drives or robotic actuators, where torque needs to be both predictable and controllable.

Table 1: Summary of BLDC specifications.

Parameter	Meaning
Rated Voltage	Voltage that the motor expects
Kv (RPM/V)	Motor speed per volt (no load) — higher = faster, lower torque
Rated Current	Max continuous current, make sure power supply can handle it
Stall Current/Torque	Important for overcoming resistance in the system
Torque Constant (Kt)	Torque per amp — higher = more torque

Comparison to DC motor

While both BLDC and brushed DC motors convert electrical energy into rotational motion, they differ significantly in design, performance, and maintenance. Brushed DC motors use mechanical brushes and a commutator to switch current through the windings, whereas BLDC motors rely on electronic commutation using sensors or sensorless controllers. As a result, BLDC motors tend to be more efficient, longer-lasting, and better suited for precise control, however, they are also more complex to drive and typically more expensive.

Pros:

- High efficiency and power-to-weight ratio
- Low maintenance and long lifespan (no brushes to wear out)
- Precise speed and torque control
- Low electrical noise

Cons:

- Needs controller to run
- More complex and costly than brushed DC motors
- Sensor alignment matters
- Can be harder to start without sensors (in sensorless versions)

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